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	Momentum of mass m_1	Momentum of mass m_2	Total linear momentum
Before collision	$p_{i1} = m_1 u_1$	$p_{i2} = m_2 u_2$	$p_i = p_{i1} + p_{i2}$ $p_i = m_1 u_1 + m_2 u_2$
After collision	$p_{f1} = m_1 v_1$	$p_{f2} = m_2 v_2$	$p_f = p_{f1} + p_{f2}$ $p_f = m_1 v_1 + m_2 v_2$

From the law of conservation of linear momentum,

Total momentum before collision (p_i) = Total momentum after collision (p_f)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad (4.46)$$

$$\text{Or } m_1 (u_1 - v_1) = m_2 (v_2 - u_2) \quad (4.47)$$

Further,

	Kinetic energy of mass m_1	Kinetic energy of mass m_2	Total kinetic energy
Before collision	$KE_{i1} = \frac{1}{2} m_1 u_1^2$	$KE_{i2} = \frac{1}{2} m_2 u_2^2$	$KE_i = KE_{i1} + KE_{i2}$ $KE_i = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$
After collision	$KE_{f1} = \frac{1}{2} m_1 v_1^2$	$KE_{f2} = \frac{1}{2} m_2 v_2^2$	$KE_i = KE_{i1} + KE_{i2}$ $KE_f = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$

For elastic collision,

Total kinetic energy before collision KE_i = Total kinetic energy after collision KE_f

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad (4.48)$$

After simplifying and rearranging the terms,

$$m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2)$$

Using the formula $a^2 - b^2 = (a + b)(a - b)$, we can rewrite the above equation as

$$m_1 (u_1 + v_1)(u_1 - v_1) = m_2 (v_2 + u_2)(v_2 - u_2) \quad (4.49)$$

Dividing equation (4.49) by (4.47) gives,